<u>Day 2</u>

Git commands and concepts:

Git, GitHub, GitLab : These are platforms used for version control and hosting code respositories.

Staging Files: use git add to prepare changes for a commit. For example: fit add app.py.

Committing changes: once added, you must commit your change with a message using git commit –m “message”

Pushing and pulling

Git push: sends your local commits to the remote repository where they can be seen.

Git pull : fetches and integrates changes from a remote repository into your local branch.

Viewing branch : git branch –all lists all available branch names in the repository.

Switching branch : git checkout allows you to switch between different branches.

Python Virtual Environments

Creating a new virtual ensures a clean server or memory space, meaning it start empty without pre-installed.

Day 3

Functional specification document

Describe what the system should do from a user perspective.

Focus on functionality and user interactions keep it non-technical for stakeholder.

Technical Specification Document

Details how the system will be implemented technically.

Includes diagram and code specification for developers.

Development quality production

Implement coding standard code reviews and testing

Use version control and smooth production deployment.

Example

Sales and marketing

Environment activation and deactivation

Document procedures for setting up or shutting down development environments.

Includes steps for configuration, licensing and termination processes.

DAY-4

DAY-4

software development lifecycle

SDLC is working on particular project and deliver.

1. Design – blue print of development.
2. Development – design plan into work.
3. Deployment – software released to user.
4. Maintenance – software kept running smoothly post-deployment.

Roles

- Functional consultant or business analyst
    - Defines the business problem they are facing.
    - Creating document named as functional document
- Developer or Programmers
    - Create basic poc and show to client.

Unit Testing

Creating some scenario's

- Tester
    - Unit Testing
    - Creating some scenario's
- Administration Team
    - Administrator of company
- Project or product manager
    - Delivery of project
- Sales & Marketing
- Human Resources
- Documentation Team
    - Documented
- Support Team
    - Supporting project

Four Documentation

1. Technical specification – what a system can do from a business.
2. Functional Specification – how the system will be implemented.
3. Test script – instruction used to test a software application or system.
4. Test results – outcome of executing test scripts.

Waterfall and model

1. Requirement
2. Design
3. Development
4. Testing
5. Deployment 6.Maintenance

Day 5

API

Bridge between frontend and backend

Front payload. Backend tends the repository.

Flask API use to construct and basic URL

Flask need to create endpoint

Advanced one compared to flask.

Assignment 1

Create python code using flask.

Day 6

To work with environment (activate or deactivate)

Deactivate – come to local terminate.

Development quality production

Sales and Marketing

(ex: detergent and doing advertisement)

Functional specification document

Technical specification document

Start development code and send it for small project,

requirement easy to practice, sequential model.

Day 7

SQL- retrieve, manage, storing and plays a vital role.

Command

- Insert
- Read
- Update
- Delete
- Select

SELECT – To view and read.

SELECT*FROM tablename;

INSERT

INSERT INTO tablename;

UPDATE value where tablename;

Delete – particular data, drop – entire data, truncate – more faster than delete.

Distinct(unique data) , alter, update, rename.

Operators - <= or >=. Or operator is reverse the condition

Empty values

Min max avg count

Json web token

Insert

Example

INSERT INTO customers (name, email)

VALUES (‘john doe’,);

Select

Example

Select * from table_name;

Left join

Right join

Day 8

Password Hashing – converting a password into fixed unreadable value using a mathematical function.

But hashed password are not converted to it original form again.

Ex:

Jansi – 12345

Jansi – 78901

Encoding and decoding encryption both will be converted back to its original form.

Slow secured password –Bcrypt

Salt, random value.

Pip install bcrypt

Import bcrypt

Convert password into bcrypt – Bcrypt.hashpw

String to bytes

Gensalt()

Methods

Strip() – removes the unwanted empty spaces from both the sides. “jansivaji”

Join() – join elements with a string. “like” “python – “ I like python”

Replace() – text.replace. java to python

Startwith() – true or false

Endwith() Find()- index values not started return as -1.

Isdigit() – return true or false

Isalpha() – checks whether they are in alphabets.

Dictionary

Get(), keys(), values(), items(), update(), pop(), pop item(), removes last added key. Clear() add(), remove()- shows error, discard() – do not show error.

Union intersection difference.

day- 9

Completing documentation.

Completing powerpoint.

Completing assignments.